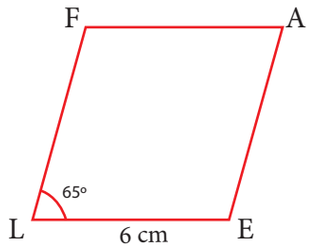
5.14.2 Construction of a rhombus when one side and one angle are given

Example 5.35

Construct a rhombus LEAF with LE = 6 cm and $\angle L = 65°$. Also find its area.



Rough diagram

Solution:

Given: LE = 6 cm and $\angle L = 65°$

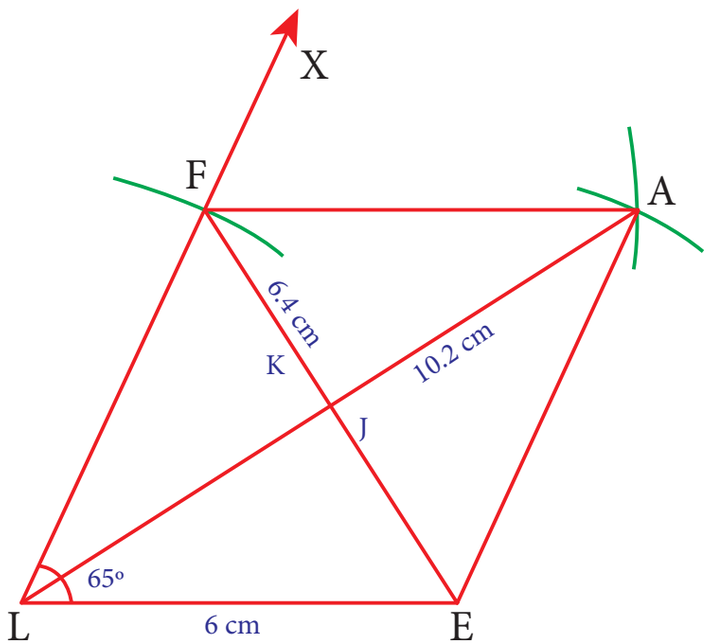


Fig. 5.54

Steps:

1. Draw a line segment LE = 6 cm.
2. At L on LE, make $\angle ELX = 65°$.
3. With L as centre draw an arc of radius 6 cm. Let it cut LX at F.
4. With E and F as centres, draw arcs of radius 6 cm each and let them cut at A.
5. Join EA and AF.
6. LEAF is the required rhombus.

Calculation of Area:

Area of rhombus LEAF $= \frac{1}{2} \times d_1 \times d_2 \; sq.units$

$= \frac{1}{2} \times 6.4 \times 10.2 = 32.64 \; sq.cm$

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